Name: _____ Class and student number: _____

Digestion of lipids and fair comparisons

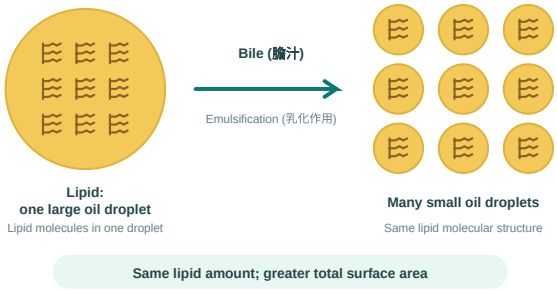
Read A from the start and keep it available. Study the explanations and examples, complete B, then receive C.

Lipids in our food have large molecular structures. They cannot pass directly through the cells lining the digestive tract (消化道) to be absorbed.

X: bile (膽汁) and physical digestion (物理消化)

X is bile (膽汁), made in the liver and stored in the gall bladder (膽囊). It contains no digestive enzymes. Emulsification (乳化作用) spreads large oil droplets into small droplets without changing lipid molecules. For the same total amount of lipid, total surface area increases, giving lipase (脂肪酶) more surface to act on.

01 • Physical digestion (物理消化): emulsification (乳化作用), with no change in molecular structure



Only the size of the oil droplets and how spread out they are change; lipid molecules are not broken down. Small droplets increase the total surface area. The diagram does not show the real sizes.

Y: lipase (脂肪酶) and chemical digestion (化學消化)

Y is lipase (脂肪酶), which speeds up lipid hydrolysis and changes chemical bonds. This course uses a complete-hydrolysis model: triglyceride (甘油三酯) + $3H_2O$ → glycerol (甘油) + 3 fatty acids (脂肪酸). The enzyme is a catalyst and is not used up.

High temperature, denaturation and fair comparison

Lipase (脂肪酶) is a protein. Boiling can cause denaturation and loss of function; cooling does not restore its action in this model. Cool boiled liquids to the common 37°C before comparison, so reaction temperature does not become another difference.

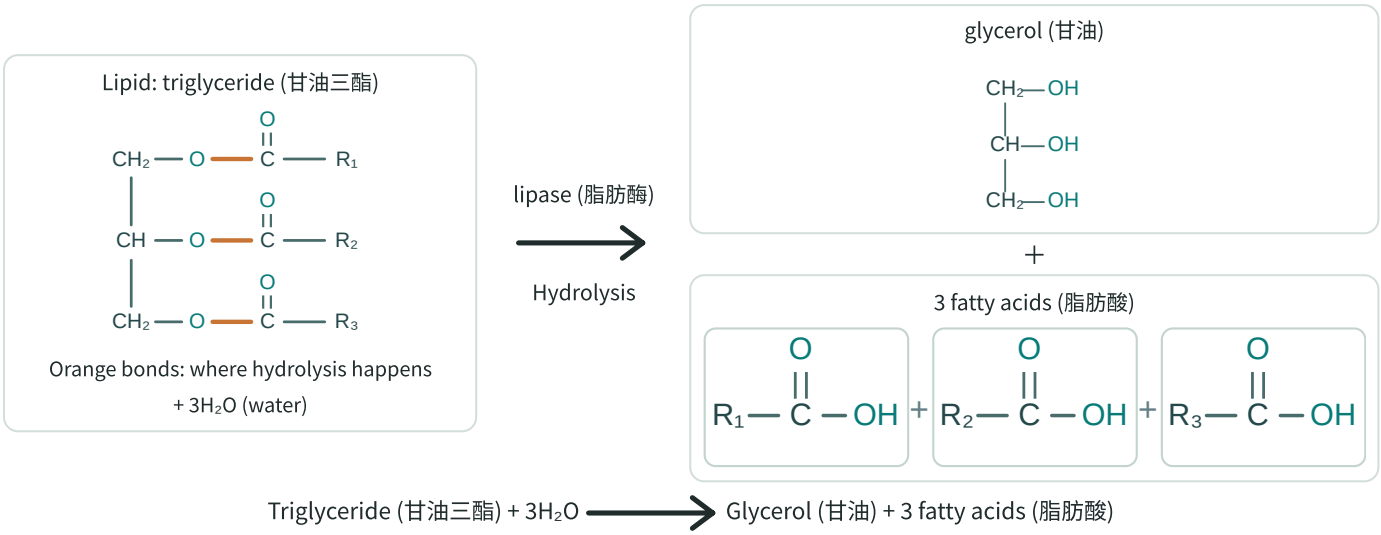
Appearance and product evidence are different

Oil layers, cloudiness and clarity are set teaching clues. They alone cannot prove glycerol (甘油) or fatty acids (脂肪酸) formed, or measure digestion speed. Real experiments need chemical tests and repeated tests.

The total activity time is 15–20 minutes. Follow the teacher’ s time limit for reading, practice, checking and correcting.

Diagrams, methods and worked examples

02 • Chemical digestion (化學消化): hydrolysis forms different molecules



R_1, R_2 and R_3 show the hydrocarbon chains of fatty acids (脂肪酸). Hydrolysis changes chemical bonds to form glycerol (甘油) and fatty acids (脂肪酸). Lipase (脂肪酶) is a catalyst and is not used up in the reaction. This diagram uses the simple complete-hydrolysis model commonly taught in this course.

The four specified sets and fixed conditions

Each set starts with the same amounts of oil and water, receives 2 mL in total, reacts at 37°C, and uses equal observation time and shaking. Recipes: control, 2 mL water; X, 2 mL X; Y, 2 mL Y; X + Y, 1 mL each. Use the researcher’ s common starting amounts, timing and shaking details; do not invent values missing from the model.

How to classify variables

The independent variable is the factor changed on purpose; the dependent variable is the result seen or measured; controlled variables are kept the same. Start with the comparison question, then find the one factor that changes.

Choose the control for the question

A control is a baseline that helps rule out other causes. It need not be free of every factor or reagent. Choose it for the actual question: adding liquid, heating, or another treatment.

Worked example: a testable prediction

“With other conditions equal, adding 2 mL X will spread large oil droplets into small droplets.” Reason: X is bile (膽汁), which can emulsify lipids (乳化脂質). Compare water with X and observe appearances. This is a worked example, not your original hypothesis. A prediction that can be observed and compared is testable; whether data support it is a separate judgement.

Worked example: classification and observation

If the question compares shaking methods, shaking is independent, the observed result is dependent, and the juice combination and other conditions stay the same. This teaches the method; it is not B’ s main-comparison answer. “Particles are at the bottom” is an observation; “they have chemically broken down” needs other evidence.

Appearance is a teaching clue: clarity alone does not prove glycerol (甘油) and fatty acids (脂肪酸) formed or measure speed. X + Y uses 1 mL each, while a single-juice set uses 2 mL. The different individual amounts cannot prove a measured combined effect or faster rate.